

ECEN 607: Advanced Analog Circuits Tech

Lab 7: Current Reference and Bandgap Circuits

Name : FirstName LastName
UIN : ***Last Three Digits**

Objectives

- To design and simulate different bandgap circuits and compare their performance.

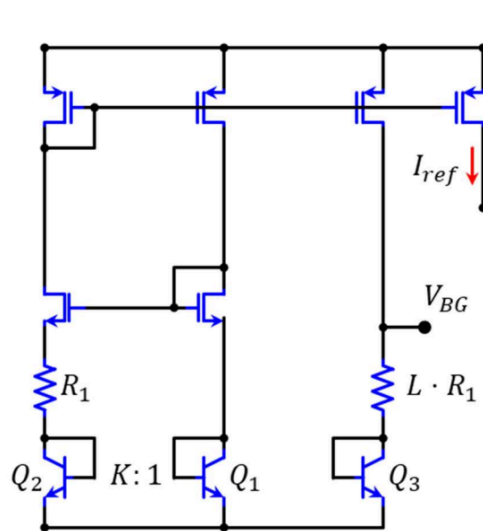


Figure 1

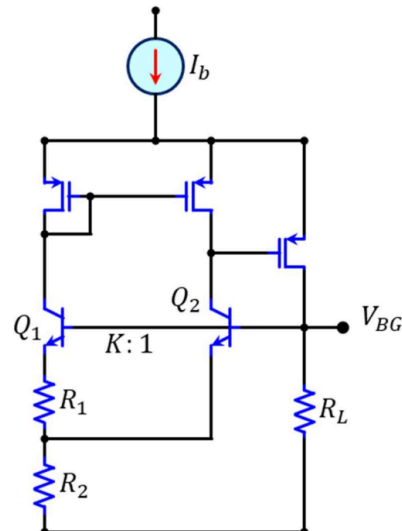


Figure 2

Prelab

- ☐ In this prelab, use 5V NFETs and PFETs and for  use the vertical devices available in our process with an emitter area of $1.0816 \mu m^2$
- ☐ Use a nominal power supply of 3V.
- ☐ Use a current density ratio of 8:1 for BJTs
- ☐ Nominal current of $50 \mu A$ in Q1 and Q2
- ☐ Voltage vs Temperature curve should be centered around 27C.
- ☐ Do not forget to show a detailed design procedure for each circuit.

Circuit 1 (Figure1):

1. Derive the expression for V_{BG} .
2. Design the circuit (including the MOS part too) and show the effect of first order temperature compensation.
3. Design a startup circuit. Mention where do you connect it and explain its functionality (not more than 4 sentences, check your lecture notes).

Circuit 2 (figure2):

4. Derive the expression for V_{BG} .
5. Design the circuit (including the MOS part too) and show the first order temperature compensation. Add a current reference circuit to generate I_b .
6. Design a startup circuit. Mention where do you connect it and explain its functionality (not more than 4 sentences).

7. Derive the approx. expression of loop gain for the two loops in the circuit; two loops? Interesting question isn't it? How to calculate loop gain?. Get an estimate of DC loop gains (take R_{ce} of BJTs to be infinite). What do you conclude? What is the need of this feedback loop?

Lab Report

1. Simulate (do for both circuits), and report the following:
 - a. Output voltage vs temperature. (-40 C to +85 C - the typical consumer electronics temperature specification).
 - b. Output voltage vs supply voltage (3 V to 4.2 V - corresponding to a commercial Li-Ion battery)
 - c. Reference current vs. temperature
 - d. Reference current vs. supply variations
 - e. Monte-Carlo simulation to specify bandgap voltage accuracy (as percentage of output voltage)
 - f. PSRR at 100Hz, 1kHz, and 100 kHz
 - g. At nominal temp, apply a ramp on Vdd (0 V to 4.2 V) and show the transient bandgap voltage (to show the working of start-up circuit)
 - h. Find maximum accuracy as percentage of output voltage.
 - i. Compare both topologies (attach one table) and indicate advantages and disadvantages.
 - j. For circuit 2 only: Show the Loop Gain (magnitude and phase plots) and report the stability metrics